Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1
The relationship between air quality and population
of cities in Kazakhstan

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Almaty

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Project goal

The primary goal of this project is to analyze the relationship between urban population sizes and air quality metrics across major cities in Kazakhstan and evaluate whether higher demographic density directly correlates with increased atmospheric pollution. To achieve this, the project implements a complete data collection and analysis pipeline by using real-world data from public Open-Meteo weather API and static website CityPopulation.

Web scraping

Imports

[1]:
```
import os
import json
import re
from time import sleep
from pathlib import Path

import unidecode
import requests
from bs4 import BeautifulSoup
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

I divided the work on the project into two parts: collecting data from web pages and APIs, and analyzing the collected data. First, I collected cities with a known population for 2026. Instead of making new requests each time, I saved all the data (including HTML and JSON) in separate files, from which I then retrieved the information. For each individual dataset, I created a CSV file.

CityPopulation

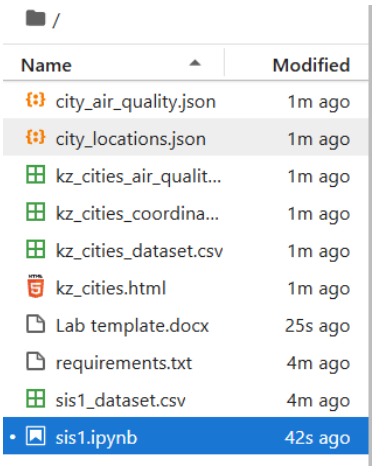
[2]:
```
WEBSITE_BASE_URL = "https://www.citypopulation.de/en/kazakhstan/cities/"
```

[3]:
```
def should_fetch(file_path: str) -> bool:
    file_path = Path(file_path)

    return not file_path.exists() or not file_path.stat().st_size > 0
```

[4]:
```
if should_fetch("kz_cities.html"):
    print("Fetching")
    try:
        with open("kz_cities.html", "w") as file:
            response = requests.get(
                url=WEBSITE_BASE_URL,
            )
            response.raise_for_status()
            file.write(response.text)
    except Exception as e:
        print(e)
print("Reading from local file")
```

```
Fetching
Reading from local file
```



Geocoding API

After receiving the list of cities and villages with their populations, I needed to find their coordinates. To do this, I used the same Open Weather API, but not for air quality, but to find the coordinates. Since the names were in Latin script, and sometimes with other variants, I needed to normalize them and convert them to the English alphabet (ASCII). For this, I used a third-party library called Unidecode. However, it didn't always translate correctly from the new Kazakh Latin script, so some city entries were deleted in the final dataset.

Retrieving cities lattitude and longtitude

[19]:
```python
def city_name_candidates(city_name_raw: str) -> list[str]:
    names = []

    main_name = re.split(r" +", city_name_raw)[0]
    if main_name:
        names.append(main_name)

    english_name = re.search(r"\[(.*)\]", city_name_raw)
    if english_name:
        names.append(english_name.group(1))

    alternate_name = re.search(r"\((.*)\)", city_name_raw)
    if alternate_name:
        names.append(alternate_name.group(1))

    for name in names:
        name = name.strip("")

    seen, out = set(), []
    for name in names:
        for name_variant in (name, unidecode.unidecode(name)):
            if name_variant not in seen:
                seen.add(name_variant)
                out.append(name_variant)

    # print(out)
    return out
```

sis1.ipynb | kz_cities_coordinates_dataset

Delimiter: ,

	Name	latitude	longitude
1	Almaty [Alma-Ata]	43.25249	76.9115
2	Astana (Nur-Sultan, Celinograd, Aqmola)	51.1801	71.44598
3	Shymkent	42.30988	69.60042
4	Aqtóbe (Akt'ubinsk) [Aktobe]	45.01214	69.81328
5	Qaraǵandy (Karaganda)	49.80187	73.10211
6	Taraz (Dzambul, Auliye-Ata)	42.89799	71.37334
7	Óskemen (Ust'-Kamenogorsk)	50.0366	82.4942
8	Pavlodar	52.27601	76.96881
9	Atyraý (Gurjev)		
10	Semeı (Semipalatinsk) [Semey]	50.42064	80.25025
11	Aqtaý (Ševcenko) [Aktau]	48.29798	80.25371
12	Qyzylorda (Kzyl-Orda) [Kyzylorda]	44.85278	65.50917
13	Qostanaı (Kustanaj) [Kostanay]	53.21435	63.62463
14	Oral (Ural'sk)	51.24601	51.42558

Air Quality API

[27]:
```python
def fetch_city_air_quality_records(longitudes: list[float], latitudes: list[float]) -> list[dict[str, str]]:
    params = {
        "longitude": ",".join(map(str, longitudes)),
        "latitude": ",".join(map(str, latitudes)),
        "hourly": ",".join([
            "european_aqi",
            "pm10",
            "pm2_5",
            "carbon_monoxide",
            "carbon_dioxide",
            "nitrogen_dioxide",
            "sulphur_dioxide",
            "ozone",
            "methane"
        ]),
        "past_days": 3,
        "forecast_days": 0,
        "timezone": "auto"
    }
    try:
        response = requests.get(
            url=OPEN_METEO_AIR_QUALITY_BASE_URL,
            params=params
        )
        response.raise_for_status()

        data = response.json()
        return data
    except Exception as e:
        print(e)
```

For each record from the created dataset with coordinates, where the coordinates were available, I made a request to the Air Quality API to obtain data on particles, gases, and the air quality index for the past 3 days for each hour. That is, there were 72 records per populated area.

Data merging

```
[42]: clean_df.head()
```

[42]:

	time	european_aqi	pm10	pm2_5	carbon_monoxide	carbon_dioxide	nitrogen_dioxide	sulphur_dioxide	ozone	methane	Name	Status	Adm.	Native	Popı
0	2026-10-26 00:00:00	58	53.1	47.3	1531.0	501	33.1	112.2	31.0	1527.0	Almaty [Alma-Ata]	City	AAK	Алматы	
1	2026-10-26 01:00:00	58	52.3	46.6	1077.0	483	32.9	105.1	31.0	1503.0	Almaty [Alma-Ata]	City	AAK	Алматы	
2	2026-10-26 02:00:00	58	51.5	45.9	726.0	470	33.0	98.2	30.0	1485.0	Almaty [Alma-Ata]	City	AAK	Алматы	
3	2026-10-26 03:00:00	57	50.2	44.7	527.0	463	33.9	91.6	29.0	1473.0	Almaty [Alma-Ata]	City	AAK	Алматы	
4	2026-10-26 04:00:00	56	49.0	43.7	431.0	460	35.1	85.2	28.0	1467.0	Almaty [Alma-Ata]	City	AAK	Алматы	

```
[43]: clean_df.to_csv("sis1_dataset.csv", index=False)
```

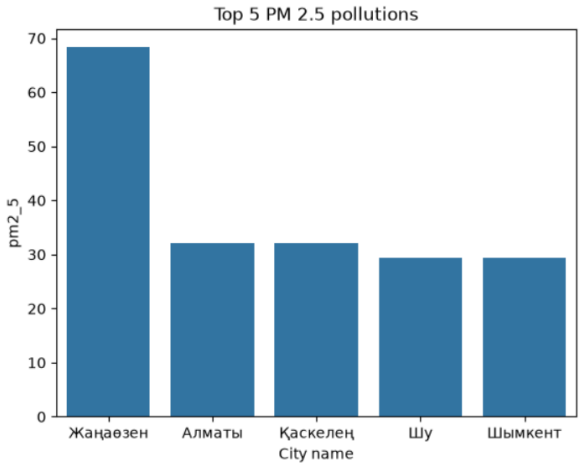
As a result, I combined 2 datasets — the populations of cities and data on their air quality. In the end, I got a dataset consisting of more than 4,000 records with hourly values of indicators for the last 3 days for each city.

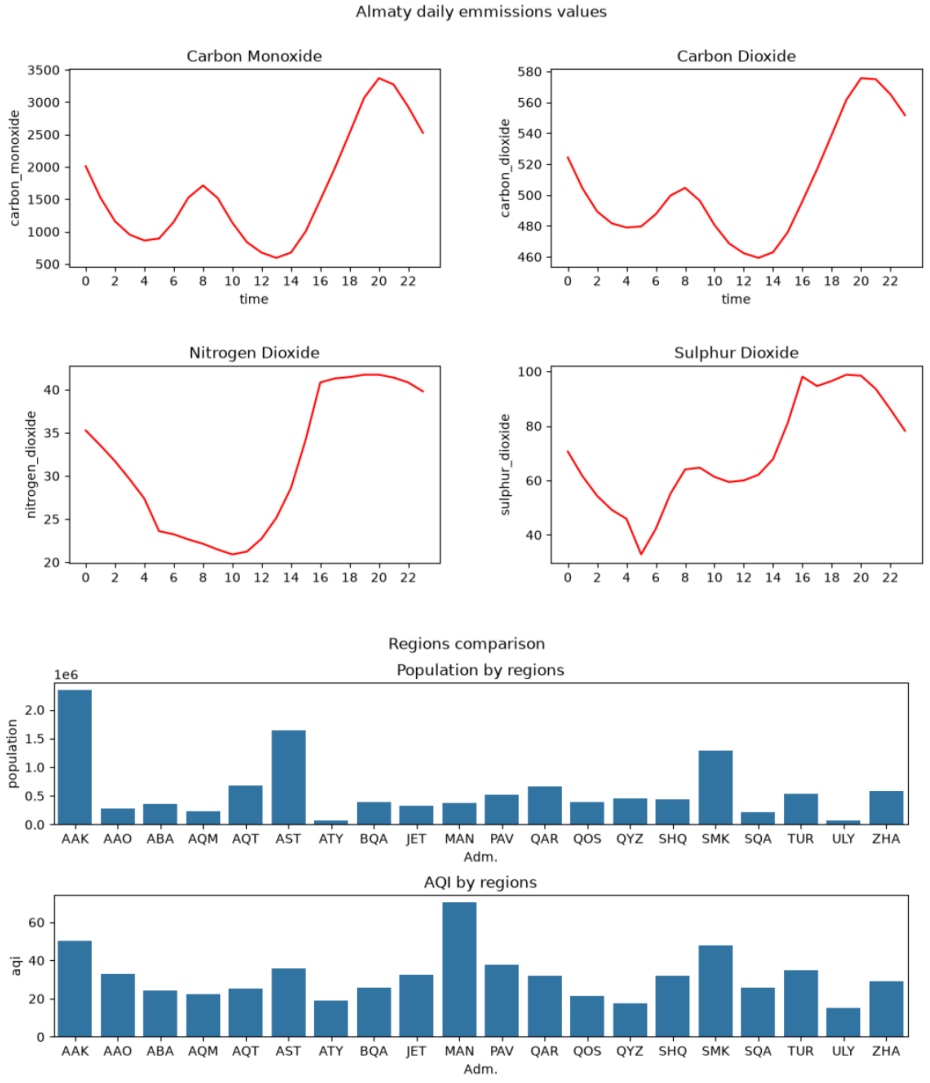
EDA

Based on the dataset I received, I conducted an initial analysis of the data, deriving 8 findings that, in my opinion, are particularly interesting and can be found in the file itself.

Based on the analysis, I concluded that there is no correlation between the population of cities in Kazakhstan and their air quality index. However, despite this, I drew conclusions about how many people are affected by polluted air, which cities are the most polluted, and how the number of particles and other gases changes throughout the day in Almaty and Zhanaozen.

```
[54]: plt.title("Top 5 PM 2.5 pollutions");
      sns.barplot(data=city_pm_stats_sorted_by_pm2_5.head(5), x="Native", y="pm2_5")
      plt.xlabel("City name");
```






```
    hue="Native",
    ax=ax[1],
    palette=palette
)
ax[1].set_title("PM 10")
ax[1].set_xlabel("Hour")
ax[1].set_xticks(range(0, 24, 2));
```

